
Null Controls Are Necessary but Not Sufficient: Auditing Acceptance Guards for Scientific Discovery and Interpretability

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Abstract

1 An automated-discovery system needs something that proposes candidates and
2 something that decides which to keep. This paper audits the deciders, reporting
3 three quantities for every acceptance guard: how often it accepts candidates built to
4 carry nothing (NAR), how often it accepts known positives (PAR), and how often
5 what it accepts survives an independent consequence test (CAR). The descriptive
6 guards in common use accept a great deal that carries nothing. A coarse-graining
7 that maps every state to zero closes perfectly on 96% of 129 chaotic systems
8 from a public benchmark, while a refutable second guard rejects it on all 129.
9 For representational correspondence the cause is structural: linear CKA gives
10 the same score to a genuine correspondence and to that map composed with a
11 random rotation, so more data does not help (NAR = 0.83 in-sample, 0.67 held
12 out, 0.80 on ESM-2 protein models). A localised difference-patching consequence
13 test refuses every null while accepting known positives (PAR = 1.00). Rejecting
14 nulls is not sufficient: on sparse-autoencoder features the correlational guard has
15 NAR = 0.00 yet disagrees with an independent intervention on more than half of
16 what it accepts (CAR@150 = 0.45 against 0.005 for matched controls). Code and
17 data: <https://real-project-aether.github.io/project-aether/>.

1 Introduction

19 An automated-discovery system needs two things: something that proposes candidates, and something
20 that decides which to keep. Deciders are receiving attention, but mostly as allocators of compute
21 [Foster et al., 2026]. This paper is about the decider that runs last, the one that certifies a candidate as
22 a result. The acceptance tests that each operation’s definition suggests turn out to accept candidates
23 that carry none of the structure the operation looks for.

24 We built four discovery operations, chosen from a corpus of scientific prizes (Sec. G). Each takes
25 a theory or a system and returns a modified one. For each we wrote the acceptance test its own
26 definition suggests, the middle column of Table 1.

27 **Three of the four accept answers that are empty; the fourth fails in the opposite direction by**
28 **firing too readily.** These are not borderline cases, and in one instance the guard destroys the structure
29 the operation was meant to find. None of the four failures is visible in a demonstration on cases
30 whose answers are known.

31 Sec. 2 states the three quantities we report, and Sec. 8 places them against the sanity-check literature
32 they generalise.

33 Contributions.

- 34 1. **One audit protocol** (Sec. 2): null acceptance, positive acceptance and consequence agree-
 35 ment, with the candidate distribution stated, applied unchanged across mechanisms that
 36 share no machinery.
- 37 2. **The guard finding** (Secs. 3, 4): four discovery operations for physics, four acceptance tests
 38 their own definitions suggest, and four failures, three of them of the same shape, replicated
 39 on 129 chaotic systems from the dysts benchmark [Gilpin, 2021], none of them ours,
 40 thresholds imported unchanged.
- 41 3. **The same audit in three interpretability settings** (Sec. 6): representational correspondence
 42 between language models and between **ESM-2 protein models**, and sparse-autoencoder
 43 features — which together show *two* distinct ways an acceptance test fails, only one of
 44 which the physics made visible.

45 **Scope of the concern.** A common way to validate a discovered structure is to show that it looks
 46 right: an analogy holds, a symmetry is recovered, a reduced model reproduces a trajectory, a latent
 47 direction is meaningful. We do not claim that every such check fails on empty inputs, and Sec. 6
 48 reports one that does not. We claim that a check is not entitled to the assumption, and that running a
 49 null is the cheapest way to find out.

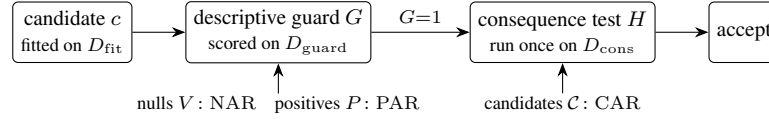


Figure 1: The audit. A guard is scored three ways: against a null suite V carrying nothing the operation seeks (NAR), a positive suite P carrying it by construction (PAR), and an independent consequence test on the candidates it accepts (CAR, over a stated distribution $\mathcal{C}$). The three partitions are disjoint; scoring a fitted candidate on its own fitting data is the most common way null acceptance is hidden.

50 2 The guard-auditing framework

51 Fig. 1 states the protocol. Write $G(c; D) \in \{0, 1\}$ for a descriptive acceptance guard applied to a
 52 candidate c given data D , and let $V = \{c_1, \dots, c_m\}$ be a *prespecified, finite* null suite: candidates
 53 carrying nothing the operation is meant to find. Of the three quantities, the first two concern nulls and
 54 answer different questions:

$$\widehat{\text{NAR}}(G; V, D) = \frac{1}{m} \sum_{c \in V} G(c; D), \quad \text{AnyNullPass}(G; V, D) = \max_{c \in V} G(c; D). \quad (1)$$

55 The **null acceptance rate** says how often the tested nulls get through; **AnyNullPass** says whether
 56 any single null defeats the guard. Neither is a supremum over an optimised family: V is a list we
 57 write down, so these are estimates against that list.

58 Null acceptance is not the only way a guard fails. A guard may reject every null and still endorse
 59 candidates that a second, independent test does not support, so we also report the **consequence**
 60 **agreement rate**

$$\text{CAR} = \Pr[H(c) = 1 \mid G(c) = 1, c \notin V], \quad (2)$$

61 where H is a prespecified consequence or intervention test. H is not ground truth: it is an independent
 62 test that can disagree with G , and the disagreement is what we measure. The two failure modes are
 63 separately diagnosable. High NAR is **null acceptance**; low CAR is **unsupported acceptance**. Sec. 6
 64 shows a guard with near-zero NAR and low CAR, so the second is not a corollary of the first.

65 CAR is conditional on $c \sim \mathcal{C}$, the distribution of candidates actually put to the consequence test,
 66 so a different proposer or selection policy gives a different number for the same guard. We state $\mathcal{C}$
 67 wherever CAR appears. In Sec. 6 it is the top K candidates by guard score within each arm, so the
 68 quantity is $\text{CAR}@K$, reported as a function of K .

Both quantities describe what a guard refuses, and a guard that refuses everything scores $\text{NAR} = 0$ while being worthless. We therefore also report the **positive acceptance rate** over a prespecified suite P of candidates constructed to carry what the operation seeks:

$$\widehat{\text{PAR}}(G; P, D) = \frac{1}{|P|} \sum_{c \in P} G(c; D). \quad (3)$$

NAR and PAR are a guard’s specificity and sensitivity, each measured against a list written down in advance.

This has a consequence for CAR that is easy to miss. CAR counts how often H fires among the candidates G accepts, so a low CAR admits two readings: G accepts candidates nothing supports, or H cannot detect support that is present. CAR alone does not separate them. The consequence test therefore needs a positive control of its own, $\text{PAR}(H; P)$, the rate at which H fires on candidates built to be detectable. A measured CAR must be read against that ceiling rather than against 1. Two consequences are used below. A matched control keeps ΔCAR interpretable even when $\text{PAR}(H; P)$ is low, because the same insensitive H scores both arms. And a positive control is only valid if its own manipulation has the effect it was built to have, which is checkable and is false in one of our arms (Sec. 6).

The rest of the paper applies this to six mechanisms: four discovery operations for physics (Sec. 4), an external benchmark (Sec. 5), and two interpretability methods (Sec. 6).

3 Four operations

Each operation turns a move the corpus records into a computation, and each needs *two* conditions to accept a candidate. Full definitions, implementations and all controls are in Appendix A.

L1, L3 and L4 in brief. L1 reads the directions along which no prediction moves, structural non-identifiability [Raue et al., 2009], as a property of a theory, and must separate probe invariance, dynamical invariance, and the local-versus-uniform test that distinguishes a gauge redundancy from a global symmetry. L3 requires what crosses between domains to be a claim the target did not have and its own dynamics can refute. L4 guards on the noise scale rather than significance. Appendix A gives all three with their controls. The audit below turns on L2, whose guards the external replication uses unchanged.

L2, coarse-graining. Where L1 asks what *parameter* detail predictions do not need, this asks what *state* detail the future does not need. A reduction Q *closes* if the coarse variables predict their own future without the fine state. We test closure twice: with a law *learned* from coarse readings alone, the situation a discovering system is in, and one *derived* by projecting the known fine law [Zwanzig, 1960] — if derived closes and learned does not, the failure was our fitter. The second guard is that the reduction must *pay*:

$$\text{Pay}(Q) = \text{clip}_{[-1,1]} \left(1 - \frac{\sum_{\phi \in \mathcal{E}} \|P(Q^\dagger Q \phi) - P(\phi)\|_\Sigma^2}{\sum_{\phi \in \mathcal{E}} \|P(\phi) - \bar{P}\|_\Sigma^2} \right), \quad \|v\|_\Sigma^2 = v^\top \Sigma^{-1} v, \quad (4)$$

where P is a prespecified observable vector, $Q^\dagger$ the adjoint of the reduction, $\bar{P}$ and Σ the mean and covariance of P over an *ensemble* $\mathcal{E}$ of states rather than one trajectory. Measured along a single trajectory the denominator nearly vanishes and every truncation scores $-\infty$, so the ensemble is not optional. Weighting by Σ^{-1} is also not optional: summing squared errors across observables of different physical units lets whichever has the larger numerical scale dominate, and the guard is then not invariant to rescaling one component. An earlier version did that. With the weighting the score is invariant under any invertible linear map of P , which is what allows one threshold to be carried between systems without retuning. Two properties are asserted in the shipped code: $\text{Pay}(I) = 1$ exactly, since the numerator vanishes, and the clip makes -1 a floor rather than a measurement. The score is fit-free, after three fitted versions failed (Sec. 4).

What proposes. Every operation above can only reject. Something must propose, and a null-space search finds only what its basis contains. We use a language model to propose and never to decide: over five episodes it separates reinterpretation from patch well above chance until identifiers are renamed, after which it is at chance — recall, not judgement (Appendix D).

Table 1: Four operations, four intuitive acceptance tests, four failures. The second guard in each row hands the system a claim it can refute; the first only describes.

	the intuitive guard	NAR	Any	null suite V (type)	what the consequence test buys
L1	remove what no probe sees	1.00	1	4 charge-carrying directions (STRUCT.-PRESERVING)	8.6×10^{-1} vs $\leq 7.6 \times 10^{-10}$
L2	the coarse variables close	0.67	1	constant, random, fast modes (DEGEN., RANDOM.)	empty cases score -0.54 , -0.50
L3	the structures match	0.33	1	6 broken-symmetry targets (CORR.-BREAKING)	4.7×10^{10}
L4	the imbalance is significant	1.00	1	2 sub-systematic imbalances (TASK-CONF.)	silent at 16σ , 30σ

4 Auditing the guards: null acceptance

Table 1 sets the four cases side by side; the rest of this section takes them in turn, and Sec. 7 reports what the operations recover when both guards are applied.

L2: the constant reduction closes by construction. If $Q(x) = q_0$ for every x , the coarse trajectory is constant and the coarse law $\dot{q} = 0$ reproduces it exactly, so the closure error is 0 for every system, trajectory and horizon. **Any acceptance test resting on closure alone accepts a reduction to nothing.** A reduction onto the *fast* modes closes for a related reason: fast modes decay to zero, so it describes nothing and evolves correctly. The experiments below measure how often a criterion of this form is used without a second condition, and what adding one costs. Of three vacuity controls, two close perfectly and only the pay guard of Eq. 4 refuses them, at -0.54 and -0.50 against a threshold of 0.50; Table 5 in Appendix A gives all three.

L1, L3 and L4. For L3, a target domain built *without* the source’s structure still passes the structural match in two seeds of six on held-out states — and in five of six scored on the states the map was fitted to, as an earlier version of this work did, so scoring in-sample overstates even this. The transferred-claim test separates them by ten orders of magnitude: the weakest genuine transfer scores 4.7×10^{10} times better than the strongest control on the refutable claim, a gap that reflects near-zero residuals on the real cases rather than a calibrated effect size. L1’s case is the sharpest because the naive guard destroys something real: five directions on the lattice are invisible to every probe. Quotienting all five, which is what “remove what no measurement can see” instructs, eliminates one redundancy and **destroys four conservation laws**, each verified conserved to 7.6×10^{-10} on the same trajectory. L4 is the opposite failure: significance is the wrong scale, and a guard firing on it would posit a particle whenever an imbalance sat many standard errors from zero. At 16σ and 30σ , both still below the systematic uncertainty, the operation stays silent. That is correct, because a calibration error of a few percent produces an imbalance of this size.

Four of our own guards were wrong, and how each was caught. The pay guard is a fourth version: earlier ones used features that could not represent the observables, scored a perfect in-sample R^2 with 37 features on 62 samples, and were dominated by scale spread. An early version of X1’s causal guard ranked a vacuous candidate above the real one; localising the intervention later removed that pathology (Appendix F). None was caught by a number looking wrong. Each was caught by a condition that cannot be satisfied vacuously. Those conditions, rather than our confidence in version four, are what we offer: a reduction keeping everything must score exactly 1.000; a guard must never rank a null above the real case; a structural score must be reported off the data it was fitted to (Table 8, Appendix C).

Two conclusions. For anyone building such systems: *a structural match is weak evidence, and weakest exactly where it is easiest to compute* — scored on the data it was fitted to it accepted a false transfer five times in six, held out, two. For evaluation practice: report a guard’s NAR against a typed null suite, and its CAR against a consequence test. None of the four failures above is visible in a demonstration on cases with known answers, because on those cases both guards agree. Sec. 6 asks whether this survives outside physics, and finds a second failure mode these four do not exhibit.

5 External replication: the guards on 129 systems we did not build

Everything in Sec. 4 was measured on systems we built. That is the obvious objection to it, so we tested the L2 guards on the `dysts` benchmark of chaotic dynamical systems [Gilpin, 2021]: 137 published systems, assembled by someone else for a different purpose, with equations we did not choose.

Protocol. The closure test, the pay score of Eq. 4 and both thresholds are *imported* from the module used for Sec. 4, not reimplemented and not retuned. Two details matter for an automated-discovery claim and an earlier version of this work got both wrong: the coarse law is fitted on trajectories *excluding* the one it is evaluated on, and acceptance rests on the **learned** law alone rather than the better of it and an oracle-derived one. Neither correction changed the null result below. Both changed the positive one: under the superseded rule the real candidate closed on one system, and under the learned-only rule it closes on none. 129 of 137 systems ran (Appendix J).

The empty reduction closes on 96% of a public benchmark. Mapping every state to zero gives a closure error of *exactly* 0.000 on 124 of 129 systems — the coarse dynamics is trivially autonomous, so it integrates perfectly. The pay score for that same reduction never exceeds -0.041 , far below the 0.50 threshold, so guard 2 refuses it on **129 of 129**. The separation is complete, and the thresholds were fixed on a different system before this benchmark was considered. Table 15 and Fig. 3 in Appendix J give the full result.

Coordinate dependence. The constant reduction lifts its coarse zero to the fine-state origin, so the result would be an artefact if closure depended on $x = 0$ being distinguished. Re-run three ways, in the original coordinates, with the origin moved well off the attractor, and as a zero-dimensional reduction with no lift at all, closure error is identical on all 38 systems where the first two are finite. The test measures closure, not our choice of origin.

Sensitivity, on a suite with ground truth. The `dysts` systems carry no ground truth, so we built a suite where the answer is fixed by construction: a system whose fast layer is slaved to its slow one as $\epsilon \rightarrow 0$, making the slow subspace the true coarse description exactly. Four dimensions, two separations and three seeds give 24 labelled systems (Table 14, Appendix J), each contributing the true reduction, the proposer’s estimate of it, and three empty candidates — the suite is the P of Eq. 3. Handed the *true* subspace the guard accepts every system ($\text{PAR} = 1.00$); handed the estimate, 0.71, so the shortfall is attributable to the proposer rather than the guard, demonstrated against ground truth rather than inferred. The vacuity finding reappears with ground truth attached: 47 of 72 empty candidates satisfy closure ($\text{NAR} = 0.65$) against 0.00 for the pair. Tolerance is neither one number nor monotonic in separation: acceptance is lost at 0.08 rad at intermediate separations and at 0.03 at both ends. Tight separation makes any tilt visible, and weak separation leaves no headroom, so the guard is about as forgiving as the timescale gap permits.

What this does *not* show, and we report as found. The real candidate reduction closes on 0 of 129 systems. This is a fact about our proposer rather than our guard: a linear projection onto slow eigen-directions of a mean Jacobian is the wrong object for a chaotic attractor, whose invariant manifold is curved. We report it because a paper showing only the controls failing would hide the more interesting negative. On this benchmark the operation never fires.

6 Auditing the guards: three interpretability settings

Everything above is physics. The failure it describes is a property of acceptance criteria rather than of physics, so it should be visible in methods built for an unrelated purpose. We ran the same pair of guards on the two objects interpretability works with directly: a claimed correspondence between two models’ representations, and a claimed meaning for a sparse-autoencoder feature. Both use Pythia models [Biderman et al., 2023] and are scored on held-out tokens; details are in Appendix F.

X1, representational correspondence. A linear map W is fitted from one model’s residual stream to another’s, and the claim is that a direction in the first means what its image means in the second. Guard 1 is the alignment quality that is standard practice — held-out R^2 and linear CKA. Guard 2

is a *localised difference patch*: perturb model A’s residual stream at one position, carry the same perturbation through W into B, and compare the induced changes in the two models’ next-token distributions. Adding the mapped *difference*, rather than overwriting B with a mapped state, keeps B in distribution and stops the causal test from inheriting the reconstruction quality that guard 1 already measures.

Every candidate is a claim about a specific map, and is scored with the map it claims. The nulls are chosen to break the correspondence while preserving what guard 1 looks at: pairings permuted before fitting, the same map composed with a random rotation, a covariance-matched Gaussian target, a map fitted on disjoint text, and a randomly initialised target model. A sixth row permutes which donor B receives while holding the map fixed — a control on the consequence test rather than a candidate. Across 4 model-layer configurations and 1920 interventions, Table 2 gives the result.

Table 2: X1 and X3, medians across configurations. **Every non-degenerate candidate carries the same CKA as the genuine correspondence** — 0.611 for the language models, 0.654 for the protein models, identical to three decimals for 5 of 5 and 5 of 5 candidates respectively — because CKA scores the two sets of representations and never looks at the map being claimed. A correspondence and its random rotation are indistinguishable to it. The consequence test separates them at AUROC 1.00 and 1.00. Rows are generated from the result files.

candidate	type	R^2	CKA	guard 1	causal	AUROC	accepted
<i>X1: Pythia pairs, 4 configurations, 480 interventions each</i>							
the correspondence under test	real	0.82	0.61	4/4	+0.80	–	4/4
pairings permuted	corr.-breaking	-0.04	0.61	3/4	-0.00	0.995	0/4
the same map, randomly rotated	corr.-breaking	-0.96	0.61	3/4	-0.01	0.996	0/4
covariance-matched Gaussian target	randomised	-0.04	0.61	3/4	+0.00	0.995	0/4
fitted on disjoint text	task-confounded	0.65	0.61	3/4	+0.01	0.996	0/4
randomly initialised target	degenerate	0.19	0.03	0/4	-0.00	1.000	0/4
donor permuted at patch time	test control	0.82	0.61	4/4	-0.09	0.965	0/4
<i>X3: ESM-2 pairs, 3 configurations, 288 interventions each</i>							
identity map (harness check)	positive	1.00	1.00	3/3	+1.00	–	3/3
adjacent layer, same model	positive	0.73	0.76	3/3	+1.00	–	3/3
the correspondence under test	real	0.47	0.65	3/3	+1.00	–	3/3
pairings permuted	corr.-breaking	-0.02	0.65	3/3	+0.01	1.000	0/3
residues permuted within protein	corr.-breaking	0.12	0.65	3/3	+0.02	1.000	0/3
the same map, randomly rotated	corr.-breaking	-0.64	0.65	3/3	+0.01	1.000	0/3
covariance-matched Gaussian target	randomised	-0.02	0.65	3/3	+0.01	1.000	0/3
randomly initialised target	degenerate	0.39	0.21	0/3	+0.03	1.000	0/3
donor permuted at patch time	test control	0.47	0.65	3/3	-0.16	0.974	0/3

More data does not fix guard 1. Scored in-sample on eight stimuli it accepts 0.83 of the nulls; scored held out on every position it still accepts 0.67. That is not the overfitting story an earlier version of this work told, and the reason is visible in the table: CKA is computed between two sets of representations, so it is unchanged by *any* claim about how they correspond. Rotating the map by a random orthogonal matrix destroys R^2 (to -0.96) and leaves CKA exactly where it was, and a guard that accepts on either criterion therefore accepts the rotation. Guard 2 refuses every null in every configuration ($\text{NAR} = 0.00$) while accepting the genuine pair in all 4.

X3, the same audit on ESM-2 protein language models. Protein language models are used to read structure and function out of sequence, and correspondence claims between them carry the same risk. We repeat the audit on ESM-2 [Lin et al., 2023] across 3 scale pairs, 320 UniRef50 proteins of 128 residues, 864 interventions. Three things make it a cleaner test than the language-model version: every ESM-2 scale shares one 33-token amino-acid vocabulary, so the two models’ outputs are comparable without any alignment step; the model is masked rather than autoregressive, so the prediction at a residue is a claim about *that* residue and the consequence is read where the intervention was made; and UniRef50 entries are cluster representatives at 50% identity, so held-out proteins are below that identity to the fitting set by construction.

X3 also carries the positive controls that make NAR interpretable, since a guard that refuses everything scores $\text{NAR} = 0$: an identity map, which cannot fail unless the harness is broken, and an adjacent layer of the same model, which is a real correspondence that must be found. Both are accepted in

every configuration, so PAR = 1.00 while NAR = 0.00. The structural guard alone accepts 0.80 of the nulls. The failure is therefore a property of the acceptance criterion rather than of English text.

A suite of named nulls says only whether those particular candidates got through, so we also drew 24 fresh members of each of 3 families for one language and one protein pair: over 144 draws guard 1 accepts 1.00 and the pair accepts 0.00, the best null reaching 0.09 (AnyNullPass = 0).

X2, an SAE feature’s meaning. Sparse autoencoders are the standard tool for extracting candidate features from a residual stream [Bricken et al., 2023]. We train a TopK autoencoder ($8\times$, $L_0 = 32$, recovering 93–99% of activation variance depending on the arm) on one model’s. Guard 1 is the correlational reading a top-activating-examples panel supports, the enrichment of a feature’s firings on a token type over its base rate. Guard 2 ablates the feature and requires the damage to be specific, degrading prediction on that concept more than elsewhere. The vacuity class is a random direction, a scrambled feature, a correlated surrogate and a near-dead feature.

The descriptive guard behaves differently here. Its NAR over four typed controls in each of four arms is 0.00 (AnyNullPass 1: a structure-preserving surrogate gets through in one arm). A co-occurrence count has almost no fitting freedom, so the mechanism that breaks alignment and closure has little to act on. It fails the *other* way.

Selection is a construction, not a result. Features are selected as the 150 most enriched per arm out of the 605–1453 that clear the threshold (Table 11), so their passing it is true by construction. CAR is the measurable quantity, and it means something only against a matched comparison. For each selected feature we draw a control from the same arm, matched on firing rate to within 20% and with a different top concept, and score it on the selected feature’s concept.

Table 3: X2. Among features selected for strong enrichment an independent intervention supports 0.45; among firing-rate-matched controls scored on the same concepts, 0.005 ($\Delta\text{CAR} = +0.45$, 95% CI +0.29 to +0.58, positive in every arm). Enrichment therefore carries causal information a matched control does not, and is not well calibrated. The interval is a cluster bootstrap over arms and over features sharing a top token, wider than treating 405 features as independent trials. Rows are generated from the result file.

arm	paired	CAR selected	CAR matched control	ΔCAR
Pythia-160m L4	93	0.24	0.00	+0.24
Pythia-160m L8	111	0.54	0.00	+0.54
Pythia-410m L16	102	0.58	0.00	+0.58
Pythia-410m L8	99	0.42	0.02	+0.40
pooled	405	0.45 (CI 0.29–0.58)	0.00	+0.45

Table 3 gives the result. The guard carries real information: matched controls are supported at 0.005 against 0.45 for selected features, the gap is positive in every arm, and its interval excludes zero. That interval was the endpoint and the retraction condition fixed in a timestamped specification before the run. Agreement is nevertheless far from complete, and no null control would have shown this, because NAR here is 0.00.

A second intervention agrees with the first. Injecting the feature where it is silent and asking whether the concept is helped is a different claim from removing what it contributes. Of the 183 features ablation supports, activation supports 162 (0.89). It is reported beside the endpoint rather than folded into it, since the support rule was fixed before the run.

The support criterion was fixed in advance. Support requires $S_f = (\Delta L_C - \Delta L_{-C})/\text{SE}[\cdot] > 1.96$ with $\Delta L_C > 0$, set in a timestamped specification before the run. An earlier version thresholded the raw ratio $\Delta L_C/\Delta L_{-C}$, which explodes near a zero denominator and reported a higher CAR as a result.

What this changes. There are two failure modes, not one, and the four physics operations only exposed the first (Table 9).

A guard can have a low NAR, 0.00 in X2’s case, and still disagree with an independent test on much of what it accepts. Null controls detect degeneracy and overfitting; they do not detect this, which is

why CAR is reported alongside, and why it needs a matched comparison to be interpretable at all. The recommendation has two halves: report a guard’s behaviour on empty inputs, *and* report how often it agrees with a refutable test on candidates that are not empty.

7 The operations recover what they should

NAR is interpretable only if the operations accept something. With both guards applied, L1 finds five invariant directions on a lattice gauge model and splits them into one redundancy and four symmetries, whose charges drift by $\leq 7.6 \times 10^{-10}$ against the redundancy’s 8.6×10^{-1} ; L2 accepts above a separation floor between $12\times$ and $25\times$ and accepts *nothing* on the lattice, learned and derived closure agreeing to three decimals, so that negative is the physics and not our fitting; L3 accepts real cases 6/6 and matched controls 0/6; L4 recovers a hidden carrier of +1.192 per event against a withheld 1.200, and is silent when the books balance. Twenty-one engineering checks all pass. Appendix B gives the full account, including a live astrophysical controversy where the criterion is silent, so the operation applies to a narrower class of disputes than we assumed.

8 Related work

The principle is not new, and neither are several of the findings. Interpretability has the idea already, method by method. Adebayo et al. [2018] showed several saliency methods produce near-identical maps after the model’s weights are randomised. Hewitt and Liang [2019] scored probes by selectivity, accuracy on the real task minus accuracy on a control task with randomised labels. Davari et al. [2023] showed CKA can be moved without changing what a network does, half of our X1; Jacovi and Goldberg [2020] separated faithfulness from plausibility; and SHAPE [Song et al., 2026] describes reasoning traces with an instrument whose own null behaviour is unreported (Appendix C).

Deployed pipelines are built from guards. Foster et al. [2026] choose which candidates are worth executing, where a misallocation is recoverable because whatever runs is checked by its own execution. Zheng et al. [2026] carry 6,453 conjectures down to 77 items for expert review, each stage a guard and the last deciding what a mathematician’s attention is spent on. Their prompt instructs the model to check that a repair is not “vacuous, or so weakened that it no longer captures the intended conjecture”: our concern, handled by instruction rather than by a measured acceptance rate.

Benchmarks now standardise these controls. Geiger et al. [2021] formalised interchange interventions and Meng et al. [2022] made activation patching standard practice; our second guard belongs to that family. SAEBench [Karvonen et al., 2025] and MIB [Mueller et al., 2025] supply shared metrics, tasks and baselines. Neither reports how often the criteria they aggregate accept a candidate built to carry nothing. A benchmark that ranks methods by agreement with a reference answer cannot measure this, because the null candidates are not among the entries being ranked.

We contribute three things. First, one audit protocol: typed nulls scored by NAR, known positives by PAR, and a prespecified consequence test scored by CAR, applied unchanged across physical reduction, representational correspondence in two model families, and feature interpretation. Prior work reports each method’s control in its own terms. Second, the failure is located in the acceptance criterion rather than in any explanation method, which accounts for its recurrence across domains that share no machinery. Third, rejecting nulls is necessary but not sufficient (Table 9), and a null control alone does not detect the second failure mode.

9 Limitations

The corpus has one annotator. Every count in Sec. G comes from a single annotator, a language model reading each official citation under human direction, so no inter-annotator agreement statistic exists. That corpus motivates the operations and supports no claim here; we ship a re-annotation kit.

The proposers are narrow. Coarse-graining and transfer search only linear maps, and recall on the proposer’s estimate of the true subspace is 0.71 against 1.00 on the subspace itself. L4 posits a quantity of charge, not an entity; L5 was not attempted; a true reduction tilted by 0.03–0.08 radians is refused; and four of eight concept-forming Standard Model episodes carry no probe for L1 (Appendix E).

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468 A The four operations in full

469 **L1, reinterpretation.** For a theory with parameters p predicting $f(x, p)$, the directions along which
 470 no prediction moves are the null space of $J_{ij} = \partial f_i / \partial p_j$: structural non-identifiability [Raue et al.,
 471 2009], read as a property of a theory rather than a nuisance in a fit. The obvious move is to quotient
 472 that null space away. *This is wrong*: a global phase rotation is invisible to every probe exactly as a
 473 gauge transformation is, and removing both erases the conserved quantity instead of discovering it.
 474 Three tests must be kept apart — **probe invariance**, **dynamical invariance**, and the **local versus**
 475 **uniform** distinction that separates gauge redundancy from a global symmetry (Table 4). We do not
 476 claim probe invariance yields a Noether charge in general; within this constructed model the local
 477 test is what classifies one.

478 **L3, transfer, and L4, positing an entity.** For transfer, find a map carrying one domain’s states into
 479 another’s so that what the first knows becomes a claim about the second: guard 1 is the structural
 480 match, guard 2 requires what crosses to be a claim the target did not have and its own dynamics can
 481 *refute*. L4 consumes what L1’s symmetry branch produces. When a found charge fails to balance
 482 across an observed process, the missing amount describes something not yet named. Its second guard
 483 is the noise scale rather than statistical significance.

484 **L1, reinterpretation.** Exact degeneracy is the wrong test for real data: Lorentz’s aether was not
 485 unobservable in principle, only to within the precision of the day. We scale column j of the Jacobian
 486 by $|p_j|$ and row i by $1/\sigma_i$, and ask whether a direction moves by an order-unity amount without
 487 shifting any prediction past its error bar. The second guard is the locality test of Table 4.

Table 4: The locality test: one null-space computation, two different moves. This is the second guard
 for L1, and the difference between deleting a structure and discovering a conservation law.

invariant <i>point by point</i>	redundancy: quotient it	[L1]
invariant only <i>uniformly</i>	symmetry: keep it, read off a conserved charge	[L4]

488 One test, two moves, and the second is the more productive: *a distinction no probe can see, but only*
 489 *when moved everywhere at once, is a conservation law*. The implementation refuses to quotient at
 490 all when no locality test is supplied: `quotient()` raises rather than proceeding, because the failure
 491 mode it guards against is silent.

492 **L2, coarse-graining.** Closure is scored by integrating the coarse law alone and comparing with
 493 the truth, normalised by the trajectory’s own spread. The *derived* law projects the known fine law
 494 through the reduction; the *learned* law is fitted from coarse readings alone. Reporting both is what
 495 makes a negative interpretable. The pay guard is fit-free by construction, after three fitted versions
 496 failed in three different ways, none of which announced itself as a wrong number. Its hard invariant is
 497 that a reduction keeping everything must score exactly 1.000.

498 **Properties of the pay score.** Write $R = Q^\dagger Q$ for the reconstruction and $\mathcal{E}$ for the state ensemble.
 499 Four properties follow directly from Eq. 4, and the first two are asserted in the shipped code rather
 500 than only claimed here.

- 501 1. **Identity.** If $R = I$ the numerator vanishes and $\text{Pay} = 1$ exactly. This is the invariant that
 502 caught three earlier fitted versions of the guard, none of which reached it.
- 503 2. **Total loss.** If R maps every state to one point, $P(R\phi)$ is constant, the numerator becomes
 504 $\sum \|P(\phi) - P_0\|^2$ for a fixed P_0 , which is at least the denominator, so the ratio is ≥ 1 and the
 505 score is at most 0 — reaching the -1 clip when P_0 is far from $\bar{P}$. Negative values therefore

Table 5: Three vacuity controls on the slow-fast system at $\epsilon = 0.02$. Two of the three close *perfectly*; only the pay guard of Eq. 4 refuses them, far below its 0.50 threshold. Pay values are from the covariance-weighted score; the unweighted version this replaced drove both empty cases to the -1 clip, which reported saturation rather than a measurement.

control	closes	pays	refused by
constant reduction (every state to zero)	0.000	-0.54	guard 2 only
random subspace	2.167	0.18	guard 1
the fast modes instead of the slow ones	0.000	-0.50	guard 2 only

mean *worse than predicting the ensemble mean*, and the clip exists so that distance is not mistaken for a measurement.

3. **Any invertible linear map of the probes.** For $P \mapsto AP$ with A invertible, the covariance becomes $A\Sigma A^\top$, so each Mahalanobis term $v^\top \Sigma^{-1}v$ is unchanged and Pay is unchanged with it. This covers scalar rescaling, unequal per-component rescaling $P \mapsto \Lambda P$, and any change of basis. Translation $P \mapsto P + b$ is likewise absorbed, since the numerator is a difference and the denominator is measured about the mean. The score is therefore a property of the reduction and the probe *span*, not of the units or coordinates the probes are written in.

The unweighted version this replaced had none of that: it was invariant under a scalar c only, so an observable with a larger numerical scale dominated the sum, and a threshold could not be carried between systems whose probes had different units. Destroying a large-scale component then scored 0.00 while destroying a small-scale one scored 1.00 for the same amount of damage; weighted they score 0.66 and 0.67. `verify.py` asserts both this and $\text{Pay}(I) = 1$.

Charge loss is reported, never blocked. Discarding state detail can destroy a conserved quantity while still closing. Physics does this deliberately, so the mechanism states which charges a reduction loses rather than refusing it; on the lattice reduction both flavour charges are lost (drift 7.5×10^{-1} and 1.9×10^0), and it says so.

L3, transfer. The structural guard requires a fitted linear map to make the target’s probes agree with the source’s to within 5%, and is scored on states the map was *not* fitted to. The second guard pushes the source’s conserved charge through that map and requires the target’s own evolution to conserve it to 10^{-3} . Table 6 gives six seeds, each with a matched control whose symmetry is deliberately broken.

Table 6: L3 transfer across six seeds, each with a matched control whose symmetry is deliberately broken. The structure test is scored both on the 40 states the map was fitted to and on 40 disjoint states; only the held-out column is used to accept. Scoring it in-sample, as an earlier version of this work did, overstates its failure rate.

	structure, sample	in-	structure, held out (< 0.05)	charge test ($< 10^{-3}$)
real	median 0.0057 — 6/6	—	median 0.0061, worst 0.0463 — 6/6 pass	median 1.3×10^{-11} , worst 5.9×10^{-11} — 6/6 pass
control	median 0.0456 — 5/6	—	median 0.0520, best 0.0405 — 2/6 pass	median 5.2, best 2.8 — 0/6 pass

L4, positing an entity. Writing ΔQ for charge in minus charge out, the operation posits a carrier of ΔQ when $\Delta Q \neq 0$ and is silent otherwise. The second guard requires $|\Delta Q|$ to exceed the *noise scale*, not merely to sit many standard errors from zero, and it fires only above about 15% of the input charge.

B Results and controls in full

The four operations have so far appeared only as things that fail in a particular way. They also work, and NAR is only interpretable if they do — a guard pair that accepted nothing would score

536 NAR = 0 while being useless, which is the reading Table 14 was built to exclude. This section is
 537 what the operations recover when both guards are applied. Negative results are reported as found,
 538 and twenty-one engineering checks on cases whose answers we supplied all pass; full numbers are in
 539 Appendix B.

540 **L1: recovering and classifying a hidden symmetry from a supplied generator basis.** On a
 541 periodic 6×6 lattice carrying a complex two-flavour matter field and U(1) links, probed only by
 542 gauge-invariant, flavour-blind quantities, the mechanism is handed eight candidate generators and told
 543 nothing about which are real. It **finds five invariant directions** and, by the locality test, splits them
 544 into **one redundancy** and **four symmetries**, emitting four conserved charges with drift 1.8×10^{-11}
 545 to 7.6×10^{-10} against the redundancy’s would-be charge at 8.6×10^{-1} — nine orders of magnitude,
 546 at the trajectory’s own precision. At uniform order the gauge transformation and the global phase act
 547 identically; only the local test separates them, and without it all four charges are destroyed to remove
 548 one redundancy.

549 **L2, L3, L4.** The reduction is accepted at $62.9 \times$ and $25.2 \times$ separation and refused below, placing
 550 the floor between $12 \times$ and $25 \times$; on the lattice it accepts *nothing* at any level, with learned and derived
 551 closure agreeing to three decimals (1.628 against 1.626), so that negative is the physics rather than
 552 our fitting. Transfer accepts real cases 6/6 and matched controls 0/6. Positing recovers a hidden
 553 carrier of +1.192 per event against a withheld truth of 1.200, and is silent when the books balance
 554 and when the imbalance is noise.

555 Appendix B also reports what happens when L1 is applied to a live astrophysical controversy with no
 556 known answer: the criterion is *silent*, because it needs two descriptions that agree on the data and
 557 these disagree by ordinary model comparison. The operation applies to a narrower class of disputes
 558 than we assumed.

559 **Engineering checks (21/21).** Every case has an answer we supplied, so passing shows the code
 560 does what it claims. The mechanism finds the unobservable direction in a description whose concepts
 561 are never separable; quotients it away with predictions unchanged to 10^{-6} ; **declines to act** on an
 562 honest theory with nothing hidden; **rejects** a patch whose new concept moves no prediction, which is
 563 the property that stops “add a parameter” from being free; and accepts a new concept only when the
 564 residual really has structure the old language lacked.

565 **L1 in full.** The uniform null space has rank 3 of 8, so five directions are invariant; a link phase,
 566 a rescale and a single-flavour shift are rejected. The local test then isolates one redundancy (local
 567 response 2.6×10^{-5}) from four symmetries (5.1 – 6.6×10^1).

568 **L2 and L4 sensitivity.** A mechanism that accepted at every separation would be broken; this one
 569 has a measurable floor between $12 \times$ and $25 \times$. For L4, at 5% and 10% of the input charge hidden
 570 the imbalance reaches 16σ and 30σ and the operation stays silent. That is not a miss: beta decay’s
 571 energy deficit was not a 5% effect, and Pauli was not reading a marginal excess.

572 **The proposer is the limiting component, not the filter.** Across three runs, a world whose residual
 573 hides $0.4 \sin 3x$ yielded eight proposals of which two to five were accepted; a control whose residual
 574 is pure noise yielded eight proposals of which **none** was accepted, in every run. Offered the true
 575 concept directly the filter separates cleanly: $\sin 3x$ reaches 97% coverage against 68% for $\sin 2.9x$
 576 and 20% for wrong terms. The model offered $\sin x$, $\sin 2x$, x^3 and $\sin \pi x$, never $\sin 3x$.

577 B.1 Where the criterion goes silent, on real data

578 We applied L1 to a live controversy with no known answer: the rotation curve of NGC 3198 from
 579 SPARC [Lelli et al., 2016], fitted by a dark-matter halo and by MOND [Milgrom, 1983]. All three
 580 descriptions carry *zero* unobservable directions — the halo model’s softest direction still moves the
 581 curve 6.4σ under a 100% excursion, unsurprising since SPARC’s photometry was assembled to break
 582 exactly that degeneracy. **The criterion is silent.** Its signature needs two descriptions that *agree* on
 583 the data; these disagree at χ^2/N of 1.22 against 9.21, where ordinary model comparison settles it.
 584 The operation applies to a narrower class of disputes than we assumed.

C The audit protocol, stated

Eq. 1 and Eq. 2 define what we report. Four things need saying about them.

These are estimates against a list, not worst cases. An earlier version of this paper defined the quantity as a supremum, $\sup_{c \in \mathcal{V}} \Pr[G(c; D) = 1]$, while reporting fractions of controls accepted. Those disagree: for a deterministic guard the supremum is 1 as soon as any null passes. NAR is the mean we were computing all along and AnyNullPass is what the supremum was reaching for, so both are now reported and neither is called a worst case. We do not optimise over a candidate family, so no supremum claim would be warranted.

V must be enumerated and typed — and we chose it. A guard with $\text{NAR} = 0$ against a weak V has been told nothing, so we list the suite and label each member by what it preserves and destroys (Table 7).

Table 7: The null suite is typed rather than pooled: a guard that survives a random direction but not a structure-preserving one has told us something specific, and pooling them into one rate hides it.

type	preserves	destroys
DEGENERATE	nothing	all structure (constant reduction, zero map)
RANDOMISED	dimensionality	any relation to the source
CORRESPONDENCE-BREAKING	marginal statistics	the pairing between source and target
STRUCTURE-PRESERVING	the invariant itself	only its redundancy status
TASK-CONFOUNDED	a trained model’s geometry	the semantic content of the input

We report this against ourselves. X1’s suite was *enlarged after an initial run*: the first three controls gave $\text{NAR} = 0.00$, we judged them too weak, and added a trained model on unrelated text, a trained model on random tokens, and a chemistry model. X2’s selection rule likewise changed after a first attempt returned features whose top token occurred three times. Strengthening V can only raise NAR, so the changes bias against our own claim, and the direction is checkable from the shipped JSON. We report per-control results throughout so a reader can recompute NAR over any sub-suite they prefer.

How diverse is a null suite? Counting nulls overstates a suite whose mass sits on one construction: six nulls that are five variants of one degenerate construction are not six independent ways of being empty. We therefore report the effective number of types a suite exercises, $\exp$ of the entropy of its type mix [Jost, 2006], which falls toward 1 as one type dominates and reaches the nominal count only when types are balanced. Our suites give 3.79 of 4 types for X1, 2.83 of 3 for X2, and 2.59 of 3 for X3, so no headline NAR here is carried by a single construction. Song et al. [2026] use the same statistic for the reverse purpose, to measure how an LLM spreads reasoning effort over interpretations of a problem.

The vacuous repair, as a null type. Our null types are degenerate, randomised, correspondence-breaking, structure-preserving and task-confounded. Zheng et al. [2026] name a specific member of the structure-preserving family that we had not, and that generalises well beyond their setting: the *vacuous repair*. When a candidate fails, a discovery system may weaken it until it is trivially true rather than abandon it, and the weakened statement keeps the objects, the notation and the shape of the original while carrying none of its content. Their pipeline guards against this by instructing the prover to check that a repair is not “so weakened that it no longer captures the intended conjecture”.

This is the same failure as the constant reduction of Sec. 4, in a different domain. A reduction mapping every state to zero is a coarse-graining weakened until closure is trivial; a vacuously repaired conjecture is a statement weakened until proof is trivial. Both keep the form and lose the content, and both are accepted by a guard that scores form. The recipe transfers: for any operation that can weaken a candidate rather than reject it, the weakened-to-triviality version belongs in V , and its acceptance rate is the quantity to report. We recommend it as a default member of the null suite for systems that repair candidates, and note that we have not run it here, since none of our four operations repairs a failed candidate.

A descriptive instrument that states its own gap. SHAPE labels chain-of-thought traces with mathematical heuristics and tracks the *semantic space* the model is working in, then reports how reasoning effort is distributed over those spaces. Its heuristic tagger is validated against an adjudicated reference set. Its semantic-space tracker is not, and the authors say why: the spaces are trajectory-relative interpretations for which, in their words, no objective gold standard exists, so the tracker is calibrated by manual review instead. By the distinction above it is a structural score, computed on the trajectory it is describing and committing to nothing that could fail, and the paper’s structural findings rest on it. What would test it is a null suite: trajectories carrying no coherent interpretation, on which a tracker that labels whatever it is given would still return a well-formed sequence of spaces. **We ran that audit and it was inconclusive**, which we report because a failed audit is still a result about what it takes to run one. Using the authors’ own coding manuals and metric code, on 24 MATH-500 problems, we compared a model’s own trace against three typed nulls (its content units reordered, a trace for a different problem, units spliced from unrelated traces) and one positive control (two independent solutions to the same problem concatenated, which contains a genuine restart). Our annotator is a 7B model, not the 27B one they use, because that model does not fit the hardware here. At that scale the tracker takes 19 NEW decisions and 0 RETURN decisions across 120 trajectories, so every condition sits at between 1.0 and 1.2 effective spaces and nothing can separate anything: the positive control itself is only at AUROC 0.39. That is a floor effect in our annotator, not a finding about their tracker, and the null AUROCs of 0.52 to 0.59 carry no information. The audit is shipped so that anyone with the 27B annotator can run it in one command.

The two kinds of guard. A structural score $S(c, D)$ is computed on the data the candidate was fitted to; a consequence test $H_c(D_{\text{test}})$ requires the candidate to commit in advance to an outcome on information not used to fit it. The evidence here supports a conditional claim rather than a general one, and the condition is not the one we first proposed. We expected S -guards to have high NAR only *when the candidate has freedom relative to D* , so that more data would fix them. That holds for the fitted map: X1’s alignment reaches $\text{NAR} = 0.83$ at eight stimuli scored in-sample. It does not explain the held-out figure of 0.67, or 0.80 for the protein models. The residue is a second mechanism entirely: linear CKA is a function of the two sets of representations, so it is invariant to *every* claim about how they correspond, and no amount of data touches that. X2’s co-occurrence count has little freedom and no such invariance, and reaches 0.00. We do not claim that consequence tests are superior in general; we claim they fail differently, and that reporting both separates the two.

Table 8: The invariants that caught our own wrong guards. A version violating one of these is wrong however plausible its numbers look, which is why four were found rather than shipped.

a reduction that keeps everything	must score exactly 1.000
a guard	must never rank a null candidate above the real one
a structural score	must be reported on data the candidate was not fitted to

Table 9: Two failure modes. Only the first is visible in the four physics operations.

	the guard accepts. . .	seen in
VACUITY	candidates that describe nothing	L2, L3, X1
OVER-ACCEPTANCE	plausible candidates no intervention supports	X2 (600 features)

Low NAR is necessary, not sufficient. X2 is the counterexample: a guard rejecting almost every null still leaves 55% of what it accepts unsupported by an independent test (Table 9). A complete report of a guard needs NAR against a typed suite *and* CAR against a prespecified consequence test.

D A language model recalls these judgements; it does not make them

A language model can recount the Lorentz–Einstein episode, so it is fair to ask whether the operation is already available in a system that has read the histories. We put five episodes, three historical and two constructed controls, to a 20B open-weight model [OpenAI, 2025] in three conditions: **named**; **blind**, every identifier alpha-renamed with fit statistics retained; and **bare**, renamed with statistics withheld. Table 10 breaks the result out by episode.

Table 10: Accuracy at identifying which of two candidates is the reinterpretation. Twelve repeats per cell, presentation order randomised, chance 50%.

episode	named	blind	bare
dark energy [Riess et al., 1998, Perlmutter et al., 1999] vs. grey dust [Aguirre, 1999]	67%	42%	50%
relativistic mass vs. Lorentz’s aether [Lorentz, 1904, Einstein, 1905]	100%	100%	100%
quasicrystals [Shechtman et al., 1984] vs. twinning [Pauling, 1985]	100%	50%	42%
two constructed controls	84%	25%	38%
pooled, 60 trials	87%	48%	53%

Accuracy is far above chance when the model can recognise the episode ($p = 5 \times 10^{-9}$) and *indistinguishable from chance* when it cannot ($p = 0.9$ blind, 0.7 bare; named against blind, paired $p = 0.03$). On these five episodes it behaves as though recalling rather than judging. We do not claim this generalises beyond them, and renaming alone is already known to move accuracy on grade-school arithmetic [Mirzadeh et al., 2025]. One case survives anonymisation: relativity stays at 100% because $(1 - v^2)^{-1/2}$ is recognisable from its algebra whatever the identifiers are called. Withholding or supplying the statistics rescues neither, so this is not a missing-information problem.

E What probe-based discovery cannot reach

The machinery above buys concepts with measurements. Tested against eight concept-forming episodes in the Standard Model it accounts for **two**: the neutrino, forced by a continuous spectrum where a two-body decay demands a line, and gauge redundancy. **Four of the other six contain no probe at all.** Colour was invented so the Δ^{++} state would be *allowed* under Fermi statistics; charm so that flavour-changing neutral currents would *cancel*; complete generations so that gauge anomalies would sum to zero; and the Higgs below about a TeV so that WW scattering would stay unitary — which told experimenters what to build.

Every one predicted a real particle, and every one was paid for by a **theorem first**, with the experiment arriving years or decades later. A framework whose founding commitment is that a concept is paid for by an experiment has these exactly backwards. The prerequisite is not more data or a better search: *you can only have theorems if there is a formal theory to have them about*, and no discovery system we know of carries one. That, rather than the difficulty of reinterpretation, is what bounds automated physics discovery today.

F The two interpretability experiments in full

Both use models from the Pythia suite [Biderman et al., 2023], whose members share a tokenizer, which is what makes the causal comparison in X1 possible at all. Text is WikiText-2. Everything is scored on held-out tokens.

X1 setup. 4 configurations over the Pythia suite: 70m L4→160m L8, 70m L2→160m L4, 160m L8→410m L12 and 70m L4→410m L12. Members share a tokenizer, which is what makes the causal comparison possible at all; text is WikiText-2, 240 sequences of 128 tokens, the map fitted on half and everything scored on the other half. Guard 1 is ridge regression ($\lambda = 10^2$) reporting held-out R^2 about the *fitting* mean, plus linear CKA, accepting at $R^2 > 0.5$ or $CKA > 0.5$. Guard 2 samples 480 donor $\times$ position $\times$ recipient triples per configuration over 5 seeds, adds the mapped difference at a single position, and correlates the two models’ induced shifts after centring across donors.

Each candidate is scored with the map it claims. A candidate that composes the fitted map with a rotation is a claim about *that* composed map, and an earlier version scored every candidate with the real map, so the rotated and donor-permuted rows reported R^2 identical to the genuine pair. Scored correctly the rotation falls to -0.96 , while its CKA does not change.

An earlier pathology, and what removed it. We previously reported that our own causal guard preferred a vacuous candidate: correlating *uncentred* shifts ranked a random-token control at 0.70

against 0.39 for the genuine pair, and centring across donors was what fixed it. That was an artefact of the intervention rather than of the metric. The old guard overwrote the residual stream at every position with one mean-pooled donor vector, which pushed both models toward the same frequent tokens, and that shared generic response dominated the correlation. Under localised difference patching the generic response is small: the worst null now scores 0.28 uncentred against 0.67 for the weakest genuine pair. Centring is still applied, and is no longer load-bearing. We report this because the earlier claim appeared in a previous version of this work and the correction runs against us.

The earlier null suite also included ChemBERTa [Chithrananda et al., 2020] reading SMILES as a different-domain control, for which guard 2 was *undefined*: the causal test compares two next-token distributions and a chemistry model shares no vocabulary with an English one. The rebuilt suite replaces it with correspondence-breaking nulls that share the vocabulary, which is a harder test, but the limitation it exposed is real and is stated in Sec. 9: the refutable guard needs a shared output space to compare in.

X3 setup. 3 ESM-2 [Lin et al., 2023] scale pairs (8M L3→35M L6, 8M L4→150M L15 and 35M L6→150M L15) on 320 UniRef50 sequences truncated to 128 residues, half fitted and half held out, 288 interventions per configuration. Guards are *imported* from the X1 module, not reimplemented. Two candidates are positive controls rather than nulls: an identity map at the same layer, which cannot fail unless the harness is broken and is there to check exactly that, and the adjacent layer of the same model, which is a genuine correspondence the guard must find. Both are accepted in every configuration.

X2 setup. A TopK sparse autoencoder on Pythia-160m layer 8, $8\times$ expansion (6,144 features), $L_0 = 32$, recovering 93–99% of activation variance depending on the arm; 1,200 sequences, 20% held out. An L_1 penalty was tried first and would not sparsify — at the coefficient that left reconstruction usable, L_0 settled near 565 and the most frequently firing features were generic ones whose top token was simply the corpus’s most common token. TopK sets L_0 by construction.

Two measurement pitfalls we hit, both of which inflate guard 1. First, ranking features by raw firing frequency selects the always-on ones; every "most interpretable" feature it returned had *the* as its top token. Second, and less obvious: scoring a feature by the likelihood ratio $P(\text{token} \mid \text{fires})/P(\text{token})$ is degenerate when the feature captures every occurrence of a rare token, because the ratio then collapses to $N/(\text{times fired})$ and is independent of how many occurrences there were. The twelve "most selective" features it first returned had top tokens occurring 3 to 29 times in held-out text — too few to intervene on. We therefore require a candidate concept to occur at least 60 times before it is scored.

Full X2 result. Table 11 gives the selection flow. A feature is selected at enrichment > 4 ; support requires the standardised difference $S_f = (\Delta L_C - \Delta L_{-C})/\text{SE}[\cdot]$ to exceed 1.96 with $\Delta L_C > 0$. An earlier version thresholded the raw ratio $\Delta L_C/\Delta L_{-C}$, which is unstable when the denominator is near zero, where one feature scored 1644. The standardised difference replaces it. This is stricter, and the reported CAR is correspondingly lower than the ratio gave.

Table 11: X2 selection flow. Features are ranked by enrichment and the top 150 per arm are taken, so the fact that every selected feature clears the threshold is a property of the pipeline and is not reported as a result. Between 605 and 1453 features per arm clear it; we test 150. Rows are generated from the result file.

stage	160m L4	160m L8	410m L16	410m L8
all SAE features	6144	6144	8192	8192
firing rate in range	3772	1248	2515	3540
concept frequent enough	3772	1248	2508	3540
enrichment > 4	1263	605	1169	1453
selected for intervention	150	150	150	150

The negative case is the sharpest: ablating that feature *improves* prediction on the very token type its top-activating examples say it carries. A panel of activating examples would have read it as interpretable.



Figure 2: What a discovering system has to be able to do. A move changes the facts you hold, the instruments you have, or the description you use; only a change of description has a depth, and its six rungs are **regression** (L0), solving inside the description you already have; **reinterpretation** (L1), the same predictions with less structure; **retyping** (L2), changing the level or quantity it is written in; **transfer** (L3), carrying another field's language in; **new object** (L4), adding a kind of thing the language lacked; and **new move** (L5), changing what counts as a move. The three column totals are annotated prize events across all fields; the per-rung counts are depth labels, since an event may sit on more than one rung (1,041 labels across 1,547 events). Of the fourteen systems we surveyed, all predominantly operate within a predefined representational or operator language; L5 is a single event its annotator flagged uncertain.

744 **Reproduction.** `xai_cka.py` and `xai_sae.py` in the release run both experiments end to end and
745 write the tables above as JSON.

746 G The taxonomy the four operations were drawn from

747 The four operations are not arbitrary, and neither is the fact that there are four. Both come from
748 asking what awarded science actually did: we annotated **1,547 prize events** by *what the contribution*
749 *did* rather than by how it was cited. A contribution changes the **facts** you hold, the **instruments** you
750 have, or the **description** you use. Only the third has a depth, and that depth is the ladder of Fig. 2;
751 Appendix H gives the corpus, its method and its limits.

752 Two numbers set the scope (Appendix H). Across all fields **53%** of awarded events change no
753 description at all, so a framework built only around reframing would miss half its subject. Restricted
754 to the ten physics prizes that inverts: **65%** change a description. On the ladder, **L0, work inside**
755 **a fixed description, carries 42 of the 542 physics depth labels**, one in thirteen. Of the fourteen
756 systems we surveyed, none exposes an operation above it. L0 is not the shallow end; a Fields Medal
757 can sit there. Four of the six rungs are what Sec. 3 builds operations for: we do not build L0, which
758 automated discovery already does, nor L5, for the reason given in Sec. 9.

759 H The prize corpus

760 The release holds **2,327 award records** across **23 prizes** (1731–2026), of which **2,221** carry the
761 awarding body's own citation and **1,547** are annotated by what the contribution did. Existing prize
762 datasets are bibliometric [Li et al., 2019], recording who won, what they published and how it was
763 cited. They cannot say whether an advance reframed something or measured something new.

764 Table 12 gives the split. Two facts follow. First, across science most awarded work is *not* a change of
765 description: 53% of events fit neither “regression” nor “reinterpretation”. Second, in physics that
766 inverts, and two-thirds of awarded work changes the description, which is why this paper is about
767 physics.

768 The placement of fourteen automated-discovery systems on this ladder is in Appendix I.

Table 12: What awarded work changed, across all fields and restricted to the ten physics prizes. The inversion is why this paper is about physics.

	all fields	physics
changed a description	45%	65%
changed no description at all	53%	33%

Table 13: Fourteen automated-discovery systems placed on the ladder, by the operation each exposes. Everything clusters on L0; **we found none that performs L1**.

system	what it does	rung	why not higher
BACON [Langley, 1981]	rediscovers laws from tables	L0	searches a fixed term language
Eureqa [Schmidt and Lipson, 2009]	symbolic regression on motion	L0	operator set given in advance
AI Feynman [Udrescu and Tegmark, 2020]	physics-informed symbolic regr.	L0	recovers known forms; invents no term
PySR [Cranmer, 2023]	evolutionary symbolic regression	L0	fixed primitive set
LaSR [Grayeli et al., 2024]	symbolic regr. with learned library	L0/L2	abstractions are shortcuts, not retypings
DreamCoder [Ellis et al., 2021]	wake-sleep library learning	L2	names structure; no theory to reinterpret
FunSearch [Romera-Paredes et al., 2024]	LLM program search + evaluator	L0	searches a fixed language
AlphaGeometry [Trinh et al., 2024]	olympiad geometry	L0	derives inside a fixed axiom set
Robot Scientist [King et al., 2004]	hypotheses with a robot	L0 + facts	chooses experiments; ontology fixed
Coscientist [Boiko et al., 2023]	autonomous chemistry	instruments	extends reach; no description changes
AI Scientist [Lu et al., 2024]	end-to-end ML papers	L0	searches a fixed hypothesis space
AlphaFold [Jumper et al., 2021]	protein structure prediction	tractability	makes computable; does not reframe
Cascade-Correlation [Fahlman and Lebiere, 1990]	grows hidden units as needed	L0	capacity, not concepts
predicate invention [Cropper and Dumančić, 2022]	invents relations in logic programs	L4*	*open after three decades of ILP

I Automated discovery systems placed on the ladder

Table 13 gives the placement, by the *operation each system exposes* rather than by the significance of its results; the claim is about published operations rather than about limits in principle.

Table 13 clusters on L0. DreamCoder [Ellis et al., 2021] reaches L2 by naming reusable structure, and predicate invention [Cropper and Dumančić, 2022] aims at L4 and remains open after three decades of inductive logic programming. **We found none that performs L1**: no system we surveyed takes a theory that already fits and returns one that fits identically while carrying less. This is a claim about the operations these systems publish, not about what they could in principle be extended to do.

J External replication: protocol details

What was skipped and why. Of 137 systems in `dysts`, 129 ran. Eight were skipped: two abstract base classes with no concrete right-hand side, and six delay-differential systems (IkedaDelay, MackeyGlass, PiecewiseCircuit, ScrollDelay, SprottDelay, VossDelay) whose state is a function on an interval rather than a vector in $\mathbb{R}^d$, so a finite-dimensional linear reduction is not defined for them. No system was dropped after seeing its result.

Table 14: The L2 guards on a suite whose correct answer is fixed by construction. **Handed the true subspace the guard accepts every one; handed the proposer’s estimate of it, it accepts 0.71 of them.** The gap is the proposer, not the guard. Meanwhile the closure test alone admits nearly two thirds of the empty candidates, and the pair admits none.

candidate	n	accepted by the pair	
true subspace, known by construction	24	24 / 24	recall 1.00
the proposer’s estimate of it	24	17 / 24	recall 0.71
fast, random, constant	72	0 / 72	precision 1.00
<i>closure test alone</i>	72	47 / 72	NAR = 0.65

Table 15: The L2 guards on 129 chaotic systems from Gilpin [2021], dimension 3–10, with guards and thresholds imported unchanged. **The closure test accepts a reduction to nothing on 96% of the benchmark. The pay test refuses it on all of it.**

reduction	closes	median closure err.	max pay score	accepted by both
slow modes (the real candidate)	1 / 129	2.07	+0.99	1
CONTROL constant (all to zero)	124 / 129	0.000	−0.03	0
CONTROL fast modes	0 / 129	1.62	+1.00	0
CONTROL random subspace	0 / 129	2.06	+0.80	0

783 **Jacobians.** The proposer needs a linear operator. Where `dysts` ships an analytic Jacobian we use
784 it; where it does not, we take central differences of the published right-hand side, which is the same
785 operator. Systems are not excluded for lacking an analytic Jacobian.

786 **The five systems where the constant reduction does not close** are BlinkingRotlet,
787 BlinkingVortex, InteriorSquirmers, SwingingAtwood and TurchinHanski. The first three
788 are driven flows with explicit time dependence and the last two have a non-zero fixed point, so in
789 none of them is the origin invariant — the mechanism by which the empty reduction closes elsewhere
790 is simply absent.

791 **The negative is not an artifact of how hard we compressed.** Keeping half the directions is an
792 aggressive cut, so we repeated the entire sweep at $k = d - 1$ — dropping a single direction, the
793 gentlest reduction that is still a reduction. The real candidate then closes on **2 of 129** systems rather
794 than 1, and the constant reduction still closes on **124 of 129** and still pays on **none**. Both halves of
795 the result are therefore insensitive to the compression ratio: the operation almost never fires because
796 a linear subspace is the wrong object for these attractors, not because we asked for too much. At this
797 gentler cut one *fast-mode* control does survive both guards, which we report as the single case in
798 2×129 control runs where the guard pair accepted something empty.

799 **One control behaves differently here than on our own system,** and the difference is informative.
800 On the lattice, the *fast-mode* reduction closed perfectly because fast modes decay to zero. On these
801 dissipative chaotic systems it does not close at all, because the fast subspace of the mean Jacobian
802 still carries attractor motion rather than decaying to nothing. The vacuity is therefore carried entirely
803 by the constant control here. Our claim narrows accordingly: *closure alone cannot distinguish a real*
804 *reduction from a reduction to nothing*, which holds on 96% of an external benchmark, rather than the
805 stronger claim that every empty reduction closes in every setting.

806 **Reproduction.** `corpus/mechanism/external_dysts.py` in the release runs the whole sweep,
807 importing the guards from `l2_coarse.py` beside it; its output ships as `external_dysts.json`.
808 `fig_external.py` regenerates Fig. 3 from that file and *asserts* the two headline counts rather than
809 trusting them, and the release’s `verify.py` re-checks both on every run.

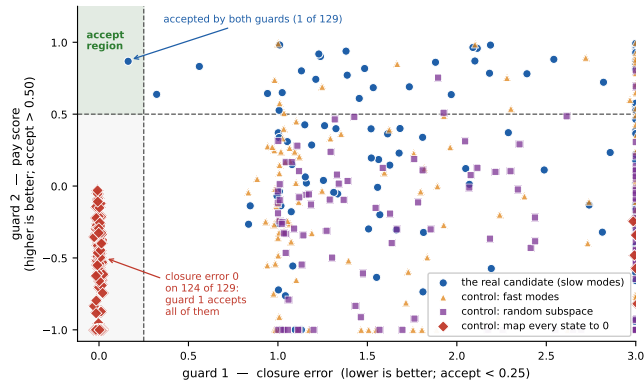


Figure 3: Both L2 guards on 129 chaotic systems from [Gilpin \[2021\]](#); one marker per (system, candidate reduction). The red stripe at closure error 0 is the reduction that maps every state to zero: it satisfies guard 1 on 124 of 129 systems and is held out of the accept region only by the pay axis, never rising above -0.030 . Closure errors that were non-finite or exceeded 3 are parked at the right edge.